

Empirical Analysis of Indoor 3D Scene Generation using Open Weight Models

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Background

- **Embodied AI (EAI)** relies on interactive multi-floor 3D simulation environments for navigation and spatial task planning [2].
- State-of-the-art 3D indoor scene generation pipelines (e.g., **MANSION** [1]), rely on closed-source cloud APIs.
- Recurring inference costs, reproducibility issues across research groups, and a lack of low-level control over structured decoding prevent access from large scale 3D scene generation.

Key Contributions

- **A fully localized, open-weight pipeline (OWP)** based on **MANSION** pipeline, executing 3D scene generation on consumer-grade hardware
- Replaces sequential cloud API calls in MANSION pipeline with a local open-weight model via Ollama [7] (Gemma-4 12B) [5].
- Achieves comparable navigability and superior physical layout validity (fewer wall collisions and boundary violations) [3].

Methodology

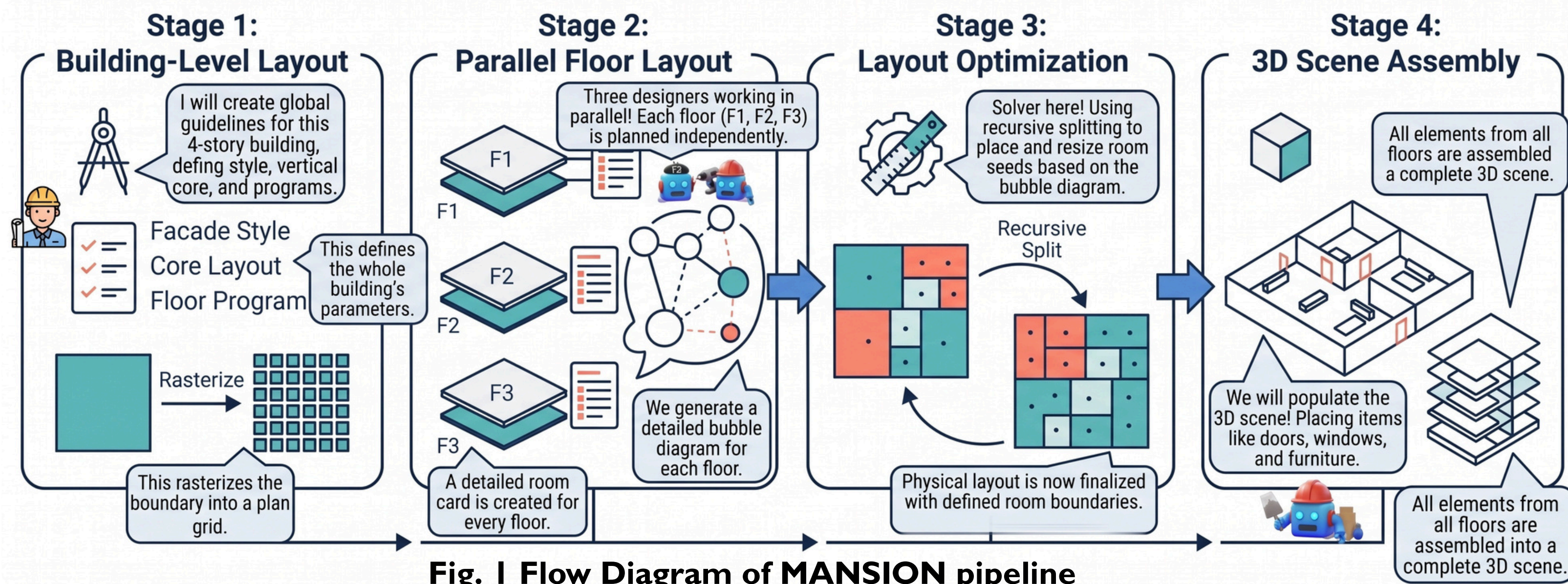


Fig. 1 Flow Diagram of MANSION pipeline

Local Hardware

Model: Gemma4:12B
CPU: Intel Corei5-12600K
GPU: NVIDIA RTX 4080
Memory: 32GB DDR5

Context Window: 32,768 tokens [6]
Output window: 8,192 tokens
T = 0.1
top_p = 0.9

Normalized Physical Scaling

To verify that lower violation counts were not simply artifacts of fewer total assets, error rates were normalized per 100 objects:

$$R_{OB} = \frac{\#OB}{N_{objects}} \times 100$$

$$R_{CN} = \frac{\#CN}{N_{objects}} \times 100$$

Table 3: Hospital Normalized Physical Quality Metrics

Metric	MANSION	OWP (Ours)
Out-of-Bound (R_{OB})	39.59	29.69
Collisions (R_{CN})	29.62	28.13

Table 4: Residential Normalized Physical Quality Metrics

Metric	MANSION	OWP (Ours)
Out-of-Bound (R_{OB})	38.16	33.56
Collisions (R_{CN})	86.47	52.35

Results



Fig. 2 Comparison of a two-story hospital facility. The top row illustrates MANSION outputs, bottom row illustrates OWP outputs

Metric	MANSION	OWP (Ours)
Objs(Small)	341(47)	64(3)
Out-of-Bound	135	19
Collisions	101	18
Reachability	94.8%	100%



Fig. 3 Comparison of a two-story residential townhouse. The top row illustrates MANSION outputs, bottom row illustrates OWP outputs

Metric	MANSION	OWP (Ours)
Objs(Small)	207(10)	149(0)
Out-of-Bound	79	50
Collisions	179	78
Reachability	94.8%	100%

Small Dataset Generation



Conclusion / Future Work

Preserved Navigability: Delivers comparable spatial planning to cloud baselines, achieving up to 100% reachability across evaluated layouts

Grounding: Normalization confirms physical validity, cutting normalized OB rates by up to 25.0% and collisions by up to 39.5%

Core Impact: Demonstrates that open-weight models are a cost-effective, reproducible, and physically sound alternative for Embodied AI simulation.

Next Steps:

- Test larger open-weight models to boost asset density and clutter without compromising structured JSON validity.
- Deploy specialized models across separate generation stages (architectural zoning, floorplan synthesis, and asset placement).

References

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- [6] S. Lundberg et al., "Guidance: Controlling Large Language Models," GitHub, 2024.
- [7] Ollama Authors, "Ollama: Run LLMs Locally," GitHub, 2023.